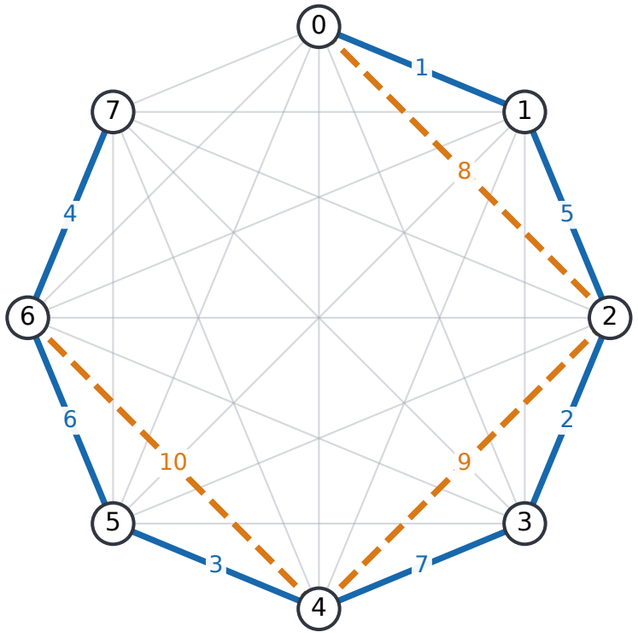
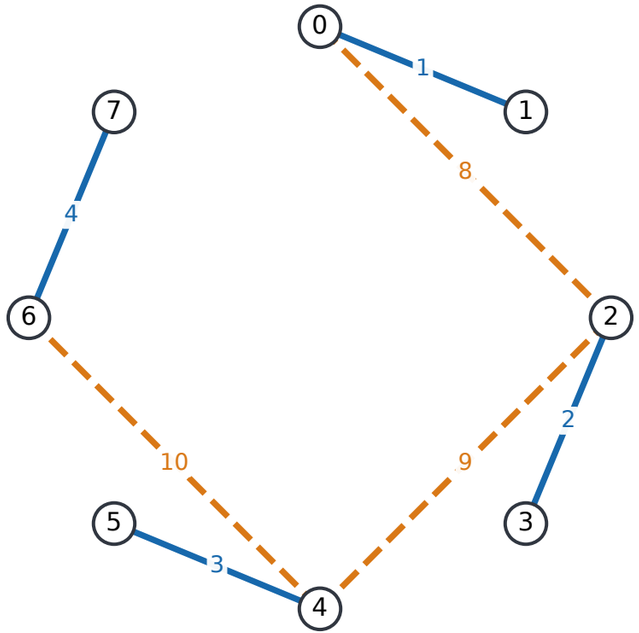
ILLUSTRATIVE WITNESS · a sampled tree can differ from the true MST and still filter safely

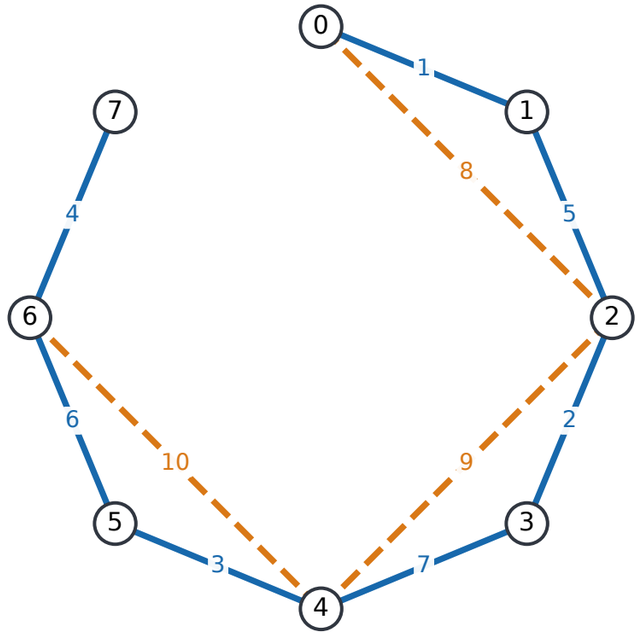
Parent graph · 28 edges
Borůvka: 3 rounds



Chosen sample H = F · 7 edges
3 non-MST chords · Borůvka: 2 rounds



F-light parent · 10 edges
18 removed · Borůvka: 3 rounds



Edges of the original unique MST Sampled forest chords outside that MST Other high chords

All edges are drawn. Important edges carry their distinct weights; other chord weights are 11–28. Vertex positions are illustrative.
An edge is removed only when its sampled-forest path has a strictly smaller maximum weight. Every original MST edge survives.
This is a chosen eight-vertex sample, not a timing input, typical random draw or performance prediction. Counts and round traces are independently checked.